

Distributive Property With Area Models Mission

Warm-up and worked example

Name: _____

Date: _____

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YOUR MISSION Break apart a multiplication problem and justify the parts with an area model.

SMART MOVE Replace the unknown and check that both sides are equal.

Math Talk

Solve 8×7 by breaking 7 into 5 and 2.

My idea:

New Words

expression	equation	variable
My meaning or picture:	My meaning or picture:	My meaning or picture:

Worked Example

Do this example with your teacher. Say what changes and what stays the same.

1. Draw an 8×7 rectangle.
2. Split it into 8×5 and 8×2 .

What the example shows: _____

Distributive Property With Area

Models Mission

Learn it and show your thinking

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YOUR MISSION Follow the learning path, then explain the big idea.

SMART MOVE Draw before asking for a rule.

[] **Step 1:** Draw an 8×7 rectangle.

[] **Step 2:** Split it into 8×5 and 8×2 .

[] **Step 3:** Record $8 \times 7 = (8 \times 5) + (8 \times 2)$.

[] **Step 4:** Split the same rectangle another way.

[] **Step 5:** Discuss why both decompositions preserve total area.

My model, drawing, or organized work:

The big idea in my own words:

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Practice lab

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YOUR MISSION Use a model, equation, or explanation for every task.

SMART MOVE Check whether your answer is reasonable.

1. Decompose 6×9 in two ways.

2. Decompose 7×12 .

3. Match area models to expressions.

4. Find and correct an incorrect decomposition.

Choose one answer and prove it another way:

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Challenge and final check

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YOUR MISSION Try an unfamiliar problem and defend your reasoning.

SMART MOVE A wrong first idea can still lead to a strong solution.

Challenge Mission

Solve 7×18 in two ways without standard long multiplication.

My plan:

My solution and proof:

Final Check

Draw and label an area model proving $8 \times 7 = 56$.

☐ I can teach it

☐ I need practice

☐ I need help

Distributive Property With Area

Models Mission

Reusable math tool

Name: _____

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YOUR MISSION Use this page to draw, model, organize, and check.

SMART MOVE Replace the unknown and check that both sides are equal.

Grid Lab

What does your drawing prove?
